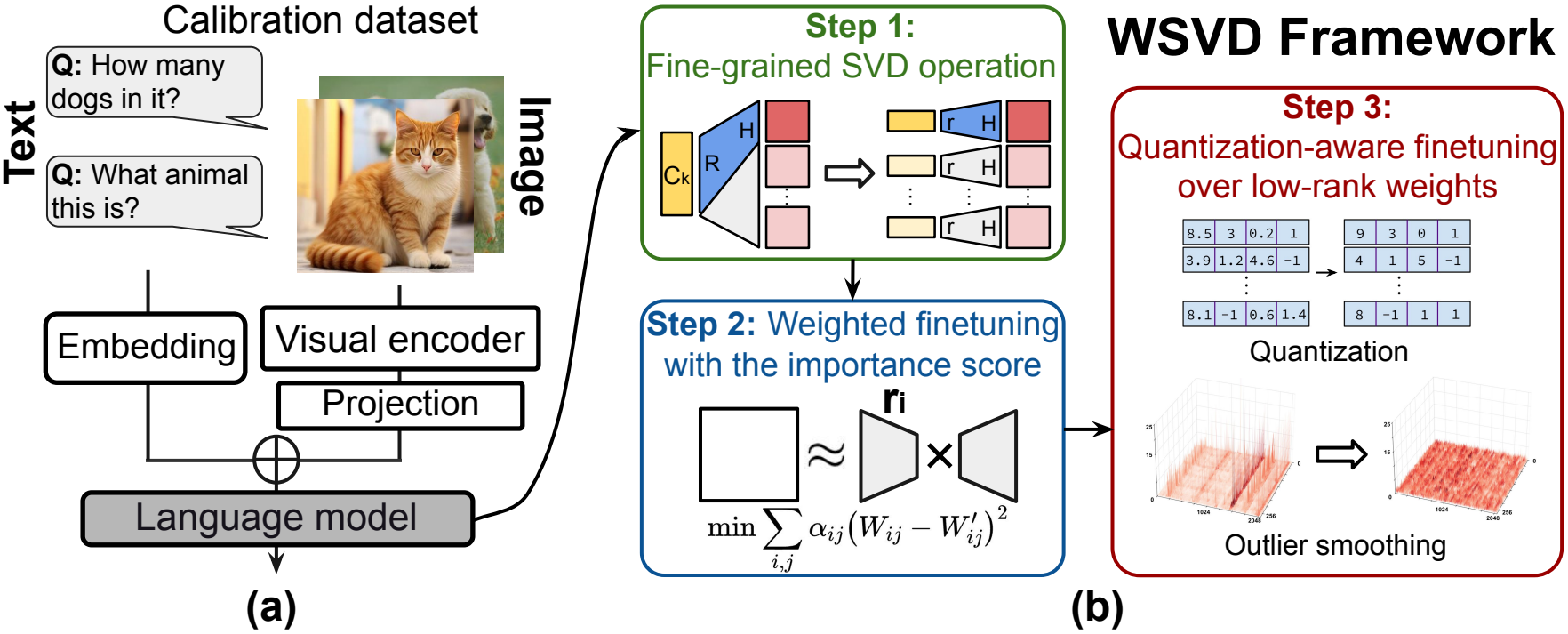
Calibration dataset
Text
Q: How many dogs in it?
Q: What animal this is?
Image
Embedding
Visual encoder
Projection
Language model
(a)
WSVD Framework
Step 1:
Fine-grained SVD operation
Ck
R
H
r
H
r
H
Step 2: Weighted finetuning with the importance score
ri
$\min \sum_{i,j} \alpha_{ij} (W_{ij} - W'_{ij})^2$
Step 3:
Quantization-aware finetuning over low-rank weights
8.5 3 0.2 1
3.9 1.2 4.6 -1
8.1 -1 0.6 1.4
9 3 0 1
4 1 5 -1
8 -1 1 1
Quantization
Outlier smoothing
(b)